Research on the Remote Sensing Change Detection of Land Cover Types in Agricultural-Forestry Transition Zone Based on CNN-Transformer Hybrid Model

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***Abstract*—In the task of land cover change detection in complex agricultural, forestry, and mountainous areas (such as karst landform areas in mountainous agricultural and forestry area), Convolutional Neural Networks (CNNs) are limited by their local receptive fields, making it difficult to effectively integrate global contextual information. Conversely, Transformer models have deficiencies in expressing fine-grained local features. These limitations make it challenging for existing methods to simultaneously address issues such as highly fragmented land features, spectral-spatial heterogeneity, and complex topographic effects. This paper proposes a hybrid model, CT-CDNet, which deeply integrates the advantages of CNNs in fine-grained local feature extraction and Transformers in global contextual dependency modeling. To systematically evaluate the model's performance, this study established a sample subset containing the aforementioned typical complex cover characteristics of the mountainous agricultural and forestry area regions based on the CLCD/px-CLCD dataset (and specifically introduced DEM data as an auxiliary input) for core experimental verification. Simultaneously, its generalization ability was tested on the LEVIR-CD and WHU-CD public benchmark datasets. The experimental results show that CT-CDNet performs outstandingly on all key evaluation indicators. For example, on the core px-CLCD mountainous agricultural and forestry area characteristic sample subset, its F1 score and Intersection over Union (IoU) reached 86.2% and 75.7%, respectively, significantly outperforming various comparative benchmark models. It also demonstrated strong competitiveness on general datasets such as LEVIR-CD (F1 of 91.9%) and WHU-CD (F1 of 89.6%). These data fully demonstrate the advancement of CT-CDNet in improving the accuracy and robustness of land cover change detection in complex agricultural, forestry, and mountainous areas, especially for challenging spectral and topographical land cover.**

Keywords-Change Detection; Deep Learning; CNN; Transformer; Hybrid Model; CLCD; px-CLCD

I. INTRODUCTION

Land cover, as a core component of the Earth's ecosystem, its spatio-temporal dynamics directly reflect the interactive impact of natural evolution and human activities. Accurate and efficient monitoring of land cover change is of irreplaceable strategic significance for regional sustainable development, ecological environment protection, rational utilization of natural resources, and the scientific formulation of related policies[1]. In particular, agricultural, forestry, and mountainous areas are often ecological function conservation areas, biodiversity hotspots, and important agricultural product supply areas. Changes in their land cover patterns profoundly affect the ecological balance and food security of the region and even larger areas. Regions like mountainous agricultural and forestry area provinces in southwestern China, with their widely distributed karst landforms, significant topographical relief, and highly fragmented land use patterns, lead to complex mosaic patterns of land cover types (such as cultivated land, forest land, grassland). Coupled with interference from factors like clouds, fog, and shadows, this poses severe challenges to remote sensing change detection technology[2]. Accurate monitoring of land cover changes in such complex areas has urgent practical needs and important research value.

Traditional remote sensing change detection methods, such as image algebra and Change Vector Analysis (CVA), can play a certain role in areas with relatively simple spectral characteristics and homogeneous land feature types[3]. However, when faced with the highly heterogeneous surface features of agricultural and forestry mountainous areas (especially karst areas like the mountainous agricultural and forestry area Plateau), frequent phenological influences, and complex lighting and shadow conditions, these methods often fail to achieve satisfactory results due to their reliance on manual thresholds and insufficient feature expression capabilities[4].

Deep learning, especially Convolutional Neural Networks (CNNs), has made significant progress in the field of change detection through its powerful local feature extraction capabilities. Models such as FCN, U-Net, and their variants have improved the accuracy of change detection through encoder-decoder or Siamese network architectures. However, the local convolution operations of CNNs limit their global contextual awareness, making it difficult to effectively distinguish land features with complex backgrounds or diverse forms of change[5]. The Transformer model, with its self-attention mechanism, performs excellently in global dependency modeling and has been applied in visual tasks (such as ViT, BIT). However, pure Transformer models may overlook local details and have high computational costs.

Given the complementary characteristics of CNNs and Transformers, hybrid models have become a research hotspot[6]. This paper focuses on the challenging problem of land cover change detection in agricultural, forestry, and mountainous areas, particularly in complex landform regions represented by mountainous agricultural and forestry area, and proposes a novel CNN-Transformer hybrid model (CT-CDNet).

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II. Related Theoretical Basis

A. Principles and Challenges of Remote Sensing Change Detection

Remote Sensing Change Detection (RSCD) aims to identify and quantify changes in surface features over time by analyzing multi-temporal remote sensing data. Its basic premise is that surface changes will cause alterations in remote sensing image features (such as spectrum, texture)[7]. However, RSCD faces many challenges, including radiometric inconsistencies between multi-temporal data, geometric registration errors, and spectral variability and confusion of complex land features (especially in agricultural and forestry areas), all of which severely restrict the accuracy of change information extraction.

B. Local Feature Perception of Convolutional Neural Networks (CNN)

Convolutional Neural Networks (CNNs) automatically learn hierarchical local features from images through core mechanisms such as local connections, weight sharing, and pooling. In change detection, CNN-based encoder-decoder (e.g., U-Net) and Siamese network architectures can effectively extract and compare fine spatial details of bi-temporal images. The advantage of CNNs lies in powerful local feature modeling, but their inherent local convolution operations make it difficult to directly capture long-range dependencies, limiting their understanding of global context[8].

C. Global Context Modeling of Transformer Models

The core of the Transformer model is the Self-Attention Mechanism, which can calculate the mutual influence between all element pairs in an input sequence, thereby effectively capturing long-range dependencies[9]. Vision Transformer (ViT) successfully applied this mechanism to visual tasks by dividing images into sequences of image patches for processing, achieving effective modeling of global image information. This provides a new way for change detection to understand scenes and correlate change areas from a global perspective. However, pure Transformers may sacrifice local detail accuracy and have higher computational and data requirements.

D. Theoretical Basis for Fusing CNN and Transformer Models

The theoretical basis for a hybrid model architecture that fuses CNNs and Transformers lies in the complementarity of their advantages. CNNs provide high-quality local features as input for the Transformer, which then performs global contextual modeling and enhancement on these features. This synergistic mechanism of "local perception - global understanding" enables the hybrid model to simultaneously capture the details of changes accurately and understand the overall patterns of change, thereby more accurately distinguishing real changes from background noise in complex remote sensing scenes (such as agricultural, forestry, and mountainous areas) and improving detection performance[10].

III. CT-CDNet: CNN-Transformer Hybrid Change Detection Network

A. Overall Model Architecture

CT-CDNet adopts a classic dual-branch Siamese network structure to process bi-temporal input images $I_1, I_2 \in \mathbb{R}^{H\times W\times C}$ (where H, W, C are the height, width, and number of channels of the image, respectively). The CT-CDNet model itself is designed as a general change detection framework and does not contain hard-coded feature fusion modules for specific geographical areas (such as mountainous agricultural and forestry area).

Its adaptability to complex regions is mainly reflected through training and testing on data containing these regional characteristics, and can be enhanced by introducing multi-source auxiliary data such as DEM as additional input channels.

B. CNN Feature Extraction Backbone

The CNN backbone network is responsible for extracting deep semantic features and multi-scale spatial information from the input bi-temporal images.

- **Network Selection:** This study selects ResNet as the basic backbone network (e.g., ResNet34 or ResNet50, pre-trained on ImageNet) because it effectively solves the gradient vanishing problem in deep network training through residual connections and has strong feature representation capabilities. The bi-temporal images $I_1$and I_2 are passed through a ResNet encoder with shared weights to obtain their respective multi-level feature maps. Assume the feature maps output from the k-th stage are $F_1^{(k)}, F_2^{(k)} \in \mathbb{R}^{H_k\times W_k\times C_k}$.
- **Feature Fusion/Differentiation:** To capture change information, the corresponding feature maps of the two time phases are processed. A simple and effective method is to calculate their absolute difference feature: $F_{diff}^{(k)} = \left|F_1^{(k)} - F_2^{(k)}\right|$.

This paper mainly adopts the feature difference method because it can more directly highlight changing areas. To retain richer contextual information, one can also consider using $F_1^{(k)}$ and $F_2^{(k)}$ (or their concatenation) simultaneously as input to the Transformer. For simplified description, F_{in} will subsequently represent the features fed into the Transformer. Deeper level features (such as the output of ResNet's stage3 or stage4) are usually selected to obtain higher-level semantic information.

C. Transformer Encoding Module

The Transformer encoding module is responsible for modeling long-range dependencies and enhancing contextual information for the features extracted by the CNN.

- **Patch Embedding:** The feature map $F_{in} \in \mathbb{R}^{H'\times W'\times C'}$ (where H′, W′, C′ are the dimensions of a deep CNN output feature map) is divided into N non-overlapping or slightly overlapping image patches, each of size P×P. The sequence length is N=(H′×W′)/P^2. Each patch is flattened and mapped to a D-dimensional feature vector through a Linear Projection layer, resulting in a feature sequence $X_p=\{x_1,x_2,\ldots,x_N\}$, where $x_i \in \mathbb{R}^D$.
- **Positional Encoding:** Since the Transformer itself cannot perceive sequence order, positional encoding $E_{pos} \in \mathbb{R}^{N\times D}$ needs to be introduced to add positional information to the feature vector of each patch: $X=X_p+E_{pos}$. Common positional encoding methods

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include fixed sinusoidal/cosinusoidal encoding or learnable positional encoding.

- **Transformer Encoder Layer:** The Transformer encoder layer is composed of multiple identical sub-layers stacked together. Each sub-layer includes a Multi-Head Self-Attention (MHSA) module and a Feed-Forward Network (FFN). A Residual Connection and Layer Normalization (LN) follow each module.

  - **Multi-Head Self-Attention (MHSA):** MHSA is the core of the Transformer. The input sequence X is first linearly mapped to Query (Q), Key (K), and Value (V) matrices: $Q=XW^Q, K=XW^K, V=XW^V$, where $W^Q, W^K, W^V \in \mathbb{R}^{D\times d_k}$ are learnable weight matrices. Attention weights are calculated by scaled dot-product: $\text{Attention}(Q,K,V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$, where d_k is the dimension of the K vector, used for scaling to prevent excessively small gradients. MHSA divides the D-dimensional Q, K, V into h heads, with each head having a dimension of $d_h=D/h$. Each head calculates attention independently, and the results are then concatenated and linearly projected again:$\text{MultiHead}(Q,K,V)=\text{Concat}(\text{head}_1,\ldots,\text{head}_h)W^O$, where $\text{head}_i=\text{Attention}(QW_i^Q, KW_i^K, VW_i^V)$, and $W_i^Q, W_i^K, VW_i^V \in \mathbb{R}^{D\times d_h}$ and $W^O \in \mathbb{R}^{D\times D}$ are also learnable weights.

  - **Feed-Forward Network (FFN):** FFN is a simple two-layer fully connected network that performs a non-linear transformation on the output of MHSA: $\text{FFN}(x)=\max(0, xW_1+b_1)W_2+b_2$, where W_1, W_2 are weight matrices and b_1, b_2 are bias terms. The output of the encoding layer, X_{out}, retains the dimensionality of the input sequence but contains richer global contextual information.

D. Change Discrimination Decoding Module

The feature sequence Xout output by the Transformer encoder needs to be converted back to a pixel-level change map.

- **Feature Reshaping and Upsampling:** First, the serialized features X_{out} are reshaped back into a feature map with spatial dimensions. Then, cascaded upsampling operations (such as transposed convolution or bilinear interpolation followed by convolution) are employed, combined with features from the shallower layers of the CNN encoder (via skip connections), to progressively restore spatial resolution and fuse multi-scale information, similar to the decoding path of U-Net. This fusion helps preserve detailed information and generate finer change boundaries.

- **Final Prediction:** The final output feature map of the decoder passes through a 1×1 convolutional layer and a Sigmoid activation function to obtain a probability map $P_{change} \in \mathbb{R}^{H\times W\times 1}$ for each pixel belonging to the changed class: $P_{change}=\text{Sigmoid}(\text{Conv}_{1\times1}(F_{decoder}))$.

E. Loss Function

Change detection is a pixel-level binary classification problem. To effectively supervise model training, this study adopts a combined loss function of Binary Cross-Entropy loss (LBCE) and Dice loss (LDice):$L_{total}=\lambda_1 L_{BCE}+\lambda_2 L_{Dice}$

where λ_1 and λ_2 are weighting coefficients to balance the two losses, usually set to $\lambda_1=\lambda_2=0.5$ or adjusted based on experience.

- **Binary Cross-Entropy Loss (BCE Loss):**

$L_{BCE} = -\frac{1}{N_{pix}}\sum_{i=1}^{N_{pix}}[y_i\log(\hat{y}_i) + (1-y_i)\log(1-\hat{y}_i)]$ where N_{pix} is the total number of pixels, $y_i \in \{0,1\}$ is the true label of pixel i (0 for unchanged, 1 for changed), and $\hat{y}_i$ is the probability predicted by the model that the pixel is changed.

- **Dice Loss:** Dice loss can effectively handle sample class imbalance problems and directly optimize the overlap between predicted results and true labels. $L_{Dice} = 1 - \frac{2\sum_{i=1}^{N_{pix}} y_i\hat{y}_i+\epsilon}{\sum_{i=1}^{N_{pix}} y_i+\sum_{i=1}^{N_{pix}}\hat{y}_i+\epsilon}$ where ϵ is a very small smoothing coefficient to prevent the denominator from being zero.

IV. Experimental Results and Analysis

A. Experimental Setup

1) Experimental Datasets and Feature Content

To comprehensively evaluate the performance of the proposed model and specifically test its adaptability in complex agricultural, forestry, and mountainous areas, this study constructed and used the following experimental data:

- **Core Experimental Data (px-CLCD mountainous agricultural and forestry area Characteristic Sample Subset):**

The core experimental data for this study were sourced from px-CLCD, a pixel-level product of the China Land Cover Dataset (CLCD).A sample subset from mountainous agricultural and forestry area provinces was specifically constructed to evaluate the model's change detection performance in complex agro-forestry mountainous regions. This subset comprehensively represents typical regional surface features: highly fragmented cropland resulting from karst topography, a highly mosaic and mixed pattern of various vegetation types (e.g., crops, forests, shrubs/grass), and mountain shadows caused by significant terrain undulation. Based on 30m Landsat imagery from corresponding years, approximately 5,000 pairs of 256×256 pixel dual-temporal image patches and their change labels were created. To enhance the model's ability to identify terrain-sensitive changes and assist in distinguishing pseudo-changes caused by shadows, SRTM DEM data were introduced as an auxiliary information channel in the experiments. This DEM data was concatenated with the multispectral bands of the optical imagery and input into the model. This core dataset was randomly divided into training (70%), validation (15%), and testing (15%) sets.

- **Public Benchmark Datasets:** To evaluate the model's generalization ability, two public building change detection benchmark datasets were additionally selected: LEVIR-CD (0.5-meter resolution, 637 image pairs) and WHU-CD (0.075/0.2-meter resolution, large sensor

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differences). Both datasets used their officially provided data splits.

2) Implementation Details and Comparative Models

- **Implementation Platform:** Experiments were based on the PyTorch deep learning framework, using the Python programming language. The hardware platform was NVIDIA GeForce RTX 3090 / A100 GPU.
- **Data Preprocessing:** All input image patches were uniformly resized to 256×256 pixels. Normalization was performed. Standard data augmentation strategies such as random flipping, rotation, and color jittering were adopted.
- **Training Parameters:** The AdamW optimizer was selected, with an initial learning rate set to 1×10−4, using a cosine annealing learning rate decay strategy. The batch size was set to 8 or 16. The number of training epochs was set to 100-200.
- **Comparative Models:** To verify the effectiveness of the proposed model, the following representative change detection models were selected for comparison: FC-EF (Fully Convolutional Early Fusion), FC-Siam-conc (Fully Convolutional Siamese Concatenation), FC-Siam-diff (Fully Convolutional Siamese Difference), UNet, SNUNet-CD (Siamese Nested U-Net for Change Detection), BIT (Bi-temporal Image Transformer), and ChangeFormer.

B. *Experimental Results and Analysis*

The following subsections will detail and analyze the experimental results obtained by applying the CT-CDNet model described in Chapter 3 and various comparative models on the aforementioned datasets and experimental conditions.

1) Performance on the px-CLCD Dataset (Agro-forestry Land Cover Change)

Quantitative Analysis: Table 1 shows the performance comparison of each model on the px-CLCD cultivated land change detection subset (key samples from the mountainous agricultural and forestry area region).

As can be seen from Table 1, on the px-CLCD dataset containing complex regional features of mountainous agricultural and forestry area, CT-CDNet achieved the best performance in all evaluation indicators. The F1-Score reached 86.2%, IoU reached 75.7%, and Kappa coefficient was 84.8%, significantly outperforming other comparative models. This indicates that CT-CDNet can effectively combine the local feature extraction capabilities of CNNs and the global context modeling capabilities of Transformers, thereby achieving more accurate change detection of cultivated land and other cover types in complex agricultural, forestry, and mountainous areas.

Table 1: Performance Comparison on the px-CLCD Cultivated Land Change Detection Subset (mountainous agricultural and forestry area Region Key Samples) (%)

Model	Precision	Recall	F1-Score	IoU	OA	Kappa
FC-EF	78.5	75.2	76.8	62.3	95.1	74.5
FC-Siam-conc	80.1	76.8	78.4	64.5	95.5	76.2
FC-Siam-diff	81.3	77.5	79.3	65.7	95.8	77.1
UNet	83.5	80.2	81.8	69.2	96.3	79.8
SNUNet-CD	85.2	82.1	83.6	71.8	96.8	81.7
BIT	84.8	83.0	83.9	72.3	96.9	82.1
ChangeFormer	86.1	83.5	84.8	73.6	97.1	83.2
CT-CDNet (Ours)	**87.5**	**84.9**	**86.2**	**75.7**	**97.5**	**84.8**

Qualitative Analysis (px-CLCD):

Qualitative Analysis (px-CLCD): In the mountainous agricultural and forestry area sample areas containing terraced fields, sloping cultivated land with tortuous boundaries and small patches, traditional CNN models such as UNet and FC series are prone to adhesion, breakage, or missed detections.Although BIT and ChangeFormer showed some improvement, they still had deficiencies in handling edge details. CT-CDNet, on the other hand, can more accurately delineate the contours of changed areas, has a higher detection rate for small cultivated land expansion or abandonment patches, and effectively suppresses pseudo-changes. In karst landform or mountainous shadow areas where land feature spectral characteristics are easily confused, CT-CDNet, by virtue of its global attention mechanism and combined with DEM auxiliary information, can refer to a wider range of contextual and terrain information for auxiliary judgment, significantly reducing misjudgments caused by local spectral similarity or shadows.

2) Performance on LEVIR-CD and WHU-CD Datasets (Building Change)

Quantitative Analysis: Figure 1 and Table 2 respectively show the performance of each model on the LEVIR-CD and WHU-CD public building change detection datasets.

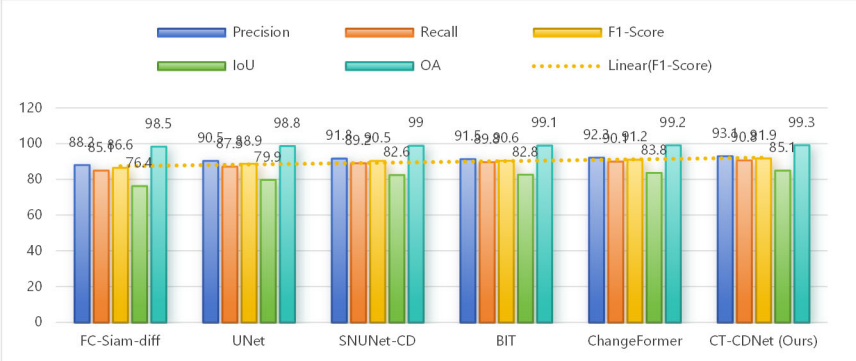


Figure 1: Performance Comparison on the LEVIR-CD Dataset (%)

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Table 2: Performance Comparison on the WHU-CD Dataset (%)

Model	Precision	Recall	F1-Score	IoU	OA
FC-Siam-diff	85.7	82.3	84.0	72.4	97.8
UNet	87.9	84.5	86.2	75.7	98.2
SNUNet-CD	89.3	86.1	87.7	78.1	98.5
BIT	89.0	86.8	87.9	78.4	98.6
ChangeFormer	90.1	87.2	88.6	79.5	98.7
CT-CDNet (Ours)	91.2	88.0	89.6	81.2	98.9

Applying CT-CDNet to the LEVIR-CD and WHU-CD datasets, its F1-Score and IoU also showed outstanding performance, as shown in Figure 1 and Table 2. These results are superior to the comparative models, fully demonstrating the effectiveness of the CT-CDNet architecture and its good generalization ability for change detection tasks with different land feature types and data qualities.

Qualitative Analysis (LEVIR-CD & WHU-CD): In the visualization results of building change detection, the building change areas detected by CT-CDNet were more complete in shape, effectively avoiding the problem of identifying large changed buildings as multiple isolated fragments. At the same time, the edges of its detection results were also smoother, clearer, and closer to the boundaries of real land features, with significantly fewer "burrs" and holes compared to some other models.

3) Ablation Experiments

To verify the effectiveness of the key components of CT-CDNet, a series of ablation experiments were conducted on the px-CLCD dataset (mountainous agricultural and forestry area characteristic sample subset), with the results shown in Table 3.

Table 3: CT-CDNet Ablation Experiment Results (px-CLCD mountainous agricultural and forestry area Characteristic Sample Subset) (%)

Model Configuration	F1-Score	IoU
CDNet (Full Model, mountainous agricultural and forestry area samples with DEM)	86.2	75.7
1. Remove Transformer module (CNN encoder + decoder, mountainous agricultural and forestry area with DEM)	82.5	70.1
2. Remove CNN backbone (Patch sequence to Transformer, mountainous agricultural and forestry area with DEM)	79.8	66.3
3. CT-CDNet (Full model, mountainous agricultural and forestry area samples without DEM	85.1	74.0
4. BCE loss only (Full model, mountainous agricultural and forestry area samples with DEM	84.5	72.8

Analysis of Table 3 Results:

- **Synergistic Effect of CNN and Transformer Modules:** Comparing the full model with configuration 1 (removing the Transformer) and configuration 2 (removing the CNN), it can be seen that both are crucial for optimal performance. Removing either part leads to a significant decrease in F1-Score and IoU, demonstrating the effective complementarity of CNN's local feature extraction and Transformer's global contextual modeling in this model.
- **Effectiveness of DEM Auxiliary Information for Complex Regions in mountainous agricultural and forestry area:** Comparing the full model (mountainous agricultural and forestry area samples with DEM) with configuration 3 (mountainous agricultural and forestry area samples without DEM), the introduction of DEM auxiliary information improved the F1-Score and IoU by 1.1% and 1.7%, respectively, when processing these samples with complex terrain features from mountainous agricultural and forestry area.This indicates that topographic factors play a positive role in distinguishing complex land cover changes in this region.
- **Effectiveness of the Combined Loss Function:** Comparing the full model with configuration 4 (BCE loss only), the combined use of BCE and Dice loss can better address the sample imbalance problem and optimize the segmentation effect of changed areas, especially reflected in the improvement of the IoU metric.

C. Comprehensive Discussion

Comprehensive quantitative, qualitative, and ablation experiments demonstrate that the CT-CDNet model achieves excellent performance across diverse tasks, including land cover change detection in agricultural, forestry, and mountainous areas, as well as general building change detection. Its main advantages stem from:

1.Deep Synergy of CNN and Transformer: The model integrates CNN's strong local detail perception with Transformer's excellent global context understanding. CNN extracts fundamental features, while Transformer, through its self-attention mechanism, models long-range dependencies to effectively distinguish between true and pseudo changes.

2.Optimized Multi-scale Feature Fusion: Through feature transfer and skip connections in the decoder, the model effectively fuses shallow features from the CNN encoder with deep semantic features from the Transformer. This ensures sensitivity to changes at different scales and a comprehensive grasp of the overall scene.

3.Adaptive Validation for Complex Agro-forestry Mountainous Features: Experiments on the px-CLCD subset, which includes complex terrain and vegetation conditions in areas like mountainous agricultural and forestry area (and incorporates DEM as auxiliary data), validated the model's capability to effectively address common challenges in agro-forestry mountainous regions, such as feature fragmentation, spectral confusion, and terrain shadows.

V. Conclusion

To address the challenges posed by complex land cover features, easily confused spectral characteristics, and the limitations of single models in balancing local details with global context, particularly in agricultural, forestry, and mountainous areas, this study proposes a hybrid change detection model, CT-CDNet, which integrates a Convolutional Neural Network (CNN) and a Transformer. This model leverages the strong local feature extraction capabilities of CNNs and the superior global context modeling ability of Transformers, aiming for the precise identification of surface

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changes within complex scenes. Experiments conducted on datasets including CLCD/px-CLCD (which contains complex mountainous agricultural and forestry samples and utilizes auxiliary DEM data), as well as the LEVIR-CD and WHU-CD datasets, demonstrate that CT-CDNet significantly outperforms various benchmark models in key performance indicators such as the F1 score and Intersection over Union (IoU) (e.g., F1: 86.2%, IoU: 75.7% on px-CLCD). These results highlight the model's high efficiency, robustness, and strong generalization ability across different change detection tasks, showing particular advantages when processing complex surface features.

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